Syntactic positional similarity modulates interference for thematic and agreement dependencies

Accounts of agreement attraction have implicated distinct underlying mechanisms: representation distortion in encoding [5,12] and similarity-based interference in retrieval [4,16]. Recent work argues that interference is best explained using both mechanisms [6,17]. The Transient Binding Model (TBM) [6] captures *item confusion*, i.e. erroneous thematic retrieval, and *item distortion*, i.e. morphosyntactic misrepresentation in encoding, under one architecture. TBM incorporates a two-layer neural network which encodes lexical items as distributed vector representations of semantic and morphological information bound to distributed representations of syntactic position. Retrieval is cued solely by the position vector of the retrieval target and, thus, the model predicts the primary determinant of interference at retrieval is the positional similarity between target and distractor encodings. However, while the thematic subject-verb dependency is highly constrained by positional similarity [1,7,15 cf. 8,14], the agreement dependency appears more permissive, exhibiting interference from a range of positionally dissimilar distractors [4,10,16]. In two experiments (E1,2), we investigate interference by varying the positional similarity of the target and distractor in (un)grammatical sentences in English, expanding the contexts modeled in [6], using a combined self-paced reading (SPR) and four alternative forced choice (4AFC) task. While we observed interference as increased target number distortion and increased item confusion in both experiments, these effects were indeed modulated by positional similarity. These results support a hybrid model of interference [6,17].

Methods: Experimental stimuli (1) realized a 2×2×2 design, crossing GRAMMATICALITY (Gram., Ungram.), DISTRACTOR NUMBER (Singular, Plural), and STRUCTURE (Double Object (DO), Prepositional Dative (PP)). In E1, the levels of DISTNUM refer to the number of the embedded subject, and in E2, the levels refer to the number of the indirect object. Thus, E1 utilizes a distractor highly similar to the target (matrix subject), whereas E2 utilizes a positionally dissimilar distractor. In each experiment, 72 participants recruited via Prolific read 48 experimental stimuli in a self-paced reading paradigm. Following each sentence, participants read a predicate and identified its subject from four alternatives (2): target or distractor lemma, whose number-marking matched (veridical) or mismatched (non-veridical) what appeared in the sentence [cf. 2,9,11,13]. Results were analyzed with Bayesian m/e regressions using `brms` [3].

E1 SPR Results (Fig. 2a)**:** We observed online evidence of agreement attraction from positionally similar distractors [cf. 10]. At the first spillover, sentences with a plural distractor were read faster than those with a singular distractor, but only for ungrammatical sentences (GRAM*DISTNUM: β = -.04, $_{95\%}$CrI = [-.09, .00]). **E1 4AFC Results** (Fig. 1a): Though we found high confusion across conditions, we observed increased item confusion in critical Ungram./Pl. conditions, and increased item distortion from plural distractors and in ungrammatical sentences (Fig. 3). Participants were more likely to misidentify the distractor as the subject (*item confusion*) in ungrammatical sentences with a plural distractor (GRAM*DISTNUM: β = -.39, $_{95\%}$CrI: [-.73, -.06]). Participants were more likely to select the non-veridical target (*item distortion*) when the distractor was plural, and in ungrammatical sentences (DISTNUM: β = -1.19, $_{95\%}$CrI = [-1.49, -.92]; GRAM: β = -1.1, $_{95\%}$CrI = [-1.4, -.82]). **E2 SPR Results** (Fig. 2b)**:** We did not observe online attraction effects from positionally dissimilar distractors. Despite numerical differences, we did not find reliable effects of the experimental factors or their interaction at any region of interest. **E2 4AFC Results** (Fig. 1b): There was high item confusion across conditions, with increased item confusion in ungrammatical sentences, and increased item distortion from plural distractors and in ungrammatical sentences (Fig. 3). Participants were more likely to misidentify the distractor as the subject in ungrammatical sentences (GRAM: β = -.28, $_{95\%}$CrI: [-.52, -.03]). Participants were more likely to select the non-veridical target when the distractor was plural, and in ungrammatical sentences (DISTNUM: β = -.53, $_{95\%}$CrI = [-.9, -.18]; GRAM: β = -1.4, $_{95\%}$CrI = [-1.77, -1.07]). There were no credible effects of STRUCT., suggesting that positional similarity refers to coarse categories of thematic roles, rather than fine-grained syntactic contexts [15].

Analyzing E1 & E2 together, we found that subject distractors engendered greater confusion in all conditions (β = -.55, $_{95\%}$CrI = [-.85, -.25]) and increased distortion when plural (β = -.74, $_{95\%}$CrI = [-1.14, -.35]), compared with indirect object distractors. These results support the predictions of the TBM, in that distractors in highly similar positions to the target elicit both greater confusion and distortion. However, the results of E2 indicate that dissimilar distractors *do* give rise to interference, but to a lesser extent. Together, the results support hybrid models of interference involving both distortion of target featural representations and erroneous retrieval of distractor encodings primarily driven by positional similarity [6,17].

(1) The chef who felt that…

the host(s)$_{E1}$ offered *the guest(s)*$_{E2}$ the menu really eagerly … [DO]

the host(s)$_{E1}$ offered the menu to *the guest(s)*$_{E2}$ really eagerly … [PP]

was/were vegetarian for the month.

(2) VEGETARIAN?

[chef]$_{target\sim veridical}$ [chefs]$_{target\sim non\text{-}veridical}$ [host$_{E1}$/guest$_{E2}$]$_{distractor\sim(non\text{-})veridical}$ [hosts$_{E1}$/guests$_{E2}$]$_{distractor\sim(non\text{-})veridical}$

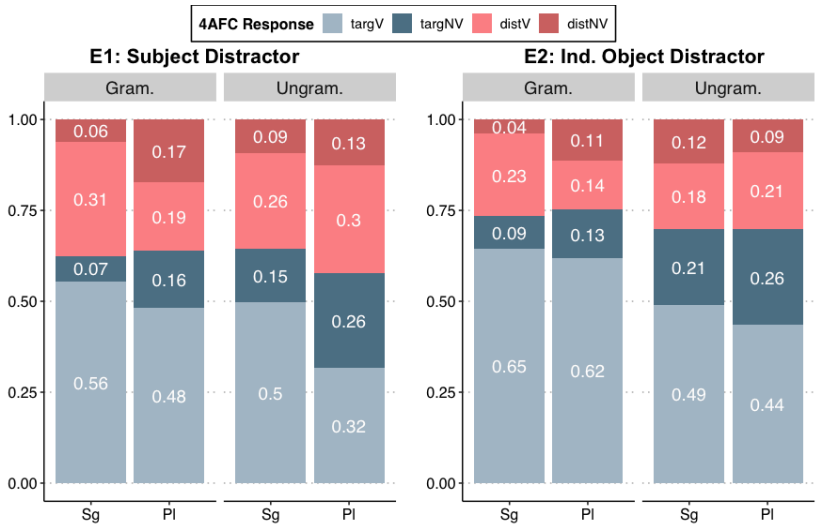

Figure 1a: E1 Proportion of Response Types by GRAM, DISTNUM

Figure 1b: E2 Proportion of Response Types by GRAM, DISTNUM

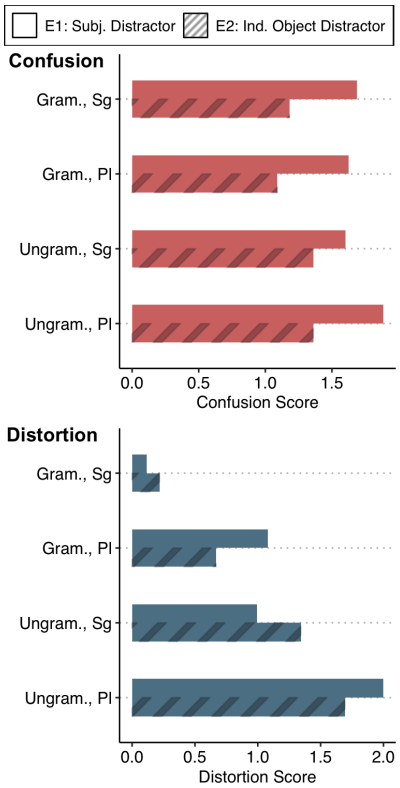

Figure 3: Confusion and Distortion by GRAM, DISTNUM across experiments

Confusion Score = log(.9/.1)-log(P(targ)/P(dist))

Distortion Score = log(.9/.1)-log(P(targV)/P(targNV))

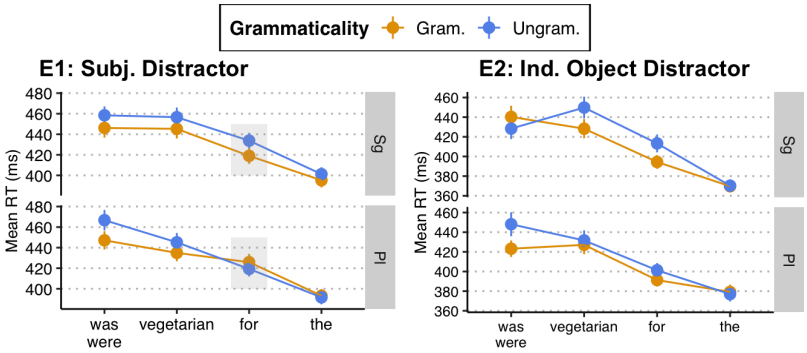

Figure 2a: E1 Mean Reading Times by GRAM, DISTNUM (error bars represent S.E.; shaded area indicates region with GRAM*DISTNUM interaction)

Figure 2b: E2 Mean Reading Times by GRAMMATICALITY, DISTRACTOR NUMBER (error bars represent S.E.)

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